

Multiobjective Optimization

The ϵ -Constraint Method and Advances

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The ϵ -Constraint Method in Multiobjective Optimization

1 Introduction and Fundamentals

- Motivation and Scalarization Approaches
- Formal Mathematical Definition

2 Classical ϵ -Constraint Method

- Operational Workflow
- Shortcomings and Limitations

3 Advanced Extensions

- AUGMECON & AUGMECON2 (Mavrotas & Florios)
- Improved, Elastic, and Adaptive Versions

4 Pareto Front Representation (GPBA Algorithms)

5 Modern Implementations & Applications

Background (Haimes, 1971):

- Primary scalarization technique alternative to Weighted Sum.
- Avoids combining distinct objectives into a subjective single function.
- Works by optimizing **one primary objective** while converting all other objectives into upper-bounded inequality constraints.

Core Logic:

- Systematically alters bound values (ϵ_k) to traverse the objective space.
- Converts a complex Multiobjective Optimization Problem (MOP) into a sequence of Single-Objective Subproblems (SOPs).
- Capable of finding non-dominated solutions in **non-convex** Pareto frontiers (unlike the classical Weighted Sum method).

General Multiobjective Optimization Problem (MOP):

$$\min_{x \in X} f(x) = (f_1(x), f_2(x), \dots, f_p(x)) \quad (p \geq 2)$$

Classical ϵ -Constraint Reformulation (P^ϵ): Select an objective $q \in \{1, \dots, p\}$ to minimize:

$$\begin{aligned} \min_{x \in X} \quad & f_q(x) \\ \text{s.t.} \quad & f_k(x) \leq \epsilon_k, \quad \forall k \in \{1, \dots, p\}, k \neq q \end{aligned}$$

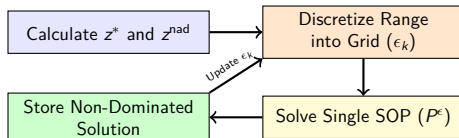
Theoretical Properties:

- **Weak Pareto Efficiency:** An optimal solution to P^ϵ guarantees at least weak Pareto optimality.
- **Strict Pareto Efficiency:** Guaranteed if the optimal solution is unique or if augmented formulations with slacks are used.

Grid Parameterization via Ideal and Nadir Points

Bounds Calculation:

- **Ideal Point (z^*):** $z_k^* = \min_{x \in X} f_k(x)$, best score attainable individually.
- **Nadir Point (z^{nad}):** $z_k^{\text{nad}} = \max_{x \in X_E} f_k(x)$, worst score over the efficient set.



Discretization Step:

$$\epsilon_k = z_k^{\text{nad}} - i_k \times \left(\frac{z_k^{\text{nad}} - z_k^*}{g_k} \right), \quad i_k \in \{0, 1, \dots, g_k\}$$

Main Limitations in Practice

1 Difficulty in Choosing ϵ Values:

- Poorly chosen bounds lead to infeasible subproblems or generate dominated / weakly efficient solutions.

2 Redundant Computations:

- Multiple ϵ -grid combinations often produce identical or redundant Pareto points due to overlapping feasible regions.

3 Non-Uniform Distribution:

- Fixed grid steps result in uneven point distributions on non-convex or highly irregular Pareto fronts.

4 Curse of Dimensionality:

- For $p > 3$, grid combinations expand exponentially ($\prod g_k$), rendering the algorithm computationally intractable.

Key Innovations:

- 1 Transforms inequality constraints into equalities using slack variables (s_k).
- 2 Introduces an augmentation term in the objective to force strict efficiency.
- 3 **AUGMECON2**: Uses slack values to dynamically skip redundant grid steps.

AUGMECON Model Formulation:

$$\begin{aligned} \max_{x \in X, s_k \geq 0} \quad & f_q(x) + \rho \sum_{k \neq q} 10^{k-1} \frac{s_k}{r_k} \\ \text{s.t.} \quad & f_k(x) - s_k = \epsilon_k, \quad \forall k \neq q \end{aligned}$$

Where $r_k = z_k^* - z_k^{nad}$ is the range of objective k , and $\rho \in [10^{-6}, 10^{-3}]$.

1. Improved / Elastic ϵ -Constraint (Ehrgott & Ruzika, 2008):

- Incorporates surplus and slack variables directly into the objective function to elasticize constraints, avoiding hard infeasibility.

2. Adaptive ϵ -Constraint (Laumanns et al., 2006):

- Dynamically resizes search boxes and varies ϵ -steps based on local landscape information, reducing unnecessary solver calls.

3. Robust Acceleration (S-AUGM / R-AUGM):

- Incorporates early exit rules and memory-efficient redundancy checks to eliminate duplicate subproblems without exact Nadir estimations (Zhang & Reimann, 2014; Nikas et al., 2020).

Grid Point Based Algorithms (Mesquita-Cunha et al., 2023)

When computing the full Pareto set is intractable, algorithms focus on finding a high-quality **representative subset** $R(N)$ evaluated across 3 dimensions:

Cardinality $\Pi(R(N))$

Coverage $\Gamma(R(N))$

Uniformity $\Delta(R(N))$

Three Dedicated GPBA Search Strategies:

- **GPBA-A (Coverage):** Minimizes maximum distance between consecutive points via dynamic interval division.
- **GPBA-B (Uniformity):** Enforces fixed minimum spacing (δ) between consecutive solutions.
- **GPBA-C (Cardinality):** Dynamically re-refines grid steps over remaining feasible regions.

Key Features & Performance Trade-offs

Variant	Redundancy Control	Target Application	Nadir Sensitivity
Classical	None	Basic Bi-objective	High
AUGMECON2	Slack-based step skip	Integer / MILP Fronts	Medium
S-AUGM	Memory / Early exit	Multi-objective Integer	Low (Insensitive)
GPBA-A	Discarded sets (D)	High Coverage Representation	Insensitive
GPBA-B	Uniformity step (δ)	Evenly Spread Fronts	Insensitive
GPBA-C	Grid refinement	Balanced Cardinality	Insensitive

Key Insights:

- Modern extensions eliminate the need for exact Nadir point calculation.
- AUGMECON2 and GPBA-C drastically reduce required MILP solver iterations.

Software Toolchains (e.g., TETREES Framework, 2026):

- Python/Gurobi-based open-source engines automating exact ϵ -constraint workflows, model presolving, and Pareto post-filtering.

Domain Applications:

- **Engineering Design:** Structural topology optimization and face milling parameter tuning.
- **Supply Chain & Logistics:** Hazardous waste location-routing, perishable inventory management.
- **Network Orchestration:** Service-to-resource allocation balancing Quality of Service (KPIs) with sustainability/ethics (KVIIs).
- **Finance:** Portfolio optimization under hard operational constraints.

Problem Definition ($n = 10$ cities)

Two cost matrices over the same 10 cities:

- d_{ij}^1 : distance (from random coordinates p^1), d_{ij}^2 : cost (from independent coordinates p^2).
- Symmetric, integer values. Independent maps $\Rightarrow$ conflicting objectives.

Decision variable: $x_{ij} = 1$ if the tour goes from city i to city j .

$$\min \quad f_1(x) = \sum_i \sum_{j \neq i} d_{ij}^1 x_{ij}, \quad f_2(x) = \sum_i \sum_{j \neq i} d_{ij}^2 x_{ij}$$

$$\text{s.t.} \quad \sum_{j \neq i} x_{ij} = 1, \quad \sum_{j \neq i} x_{ji} = 1 \quad \forall i$$

$$u_i - u_j + (n - 1) x_{ij} \leq n - 2 \quad \forall i \neq j, \quad i, j \geq 1 \quad (\text{MTZ})$$

$$x_{ij} \in \{0, 1\}, \quad 1 \leq u_i \leq n - 1$$

Why the MTZ Constraints?

- Degree constraints alone allow **several small cycles** (e.g., $0 \rightarrow 1 \rightarrow 0$ and $2 \rightarrow 3 \rightarrow 4 \rightarrow 2$), which is not a tour.
- u_i represents the **position** of city i in the tour.
- If $x_{ij} = 1$, the constraint forces $u_j \geq u_i + 1$: positions increase along the path.
- A cycle that does not contain city 0 would require positions to grow indefinitely $\Rightarrow$ infeasible.
- If $x_{ij} = 0$, the constraint is slack and has no effect.

Remark: simple and compact, but a weak formulation. For larger instances, use lazily generated subtour cuts or stronger formulations.

Subproblem and Algorithm (Exact Method)

Subproblem (f_1 primary, f_2 bounded):

$$\begin{aligned} \min_{x \in X, s \geq 0} \quad & f_1(x) - \rho \frac{s}{r_2} \\ \text{s.t.} \quad & f_2(x) + s = \epsilon \end{aligned}$$

with $\rho = 10^{-4}$ and $r_2 = z_2^{\text{nad}} - z_2^*$. The slack term breaks ties in favor of smaller f_2 (strict efficiency).

Algorithm:

- 1 **Lexicographic optimization** gives z^* and the exact nadir: $\min f_1$ then $\min f_2$ (and vice versa).
- 2 Start with $\epsilon = z_2^{\text{nad}} - 1$.
- 3 Solve the MILP and obtain (f_1^*, f_2^*) . Store the point.
- 4 **AUGMECON2-style skip:** since costs are integer, set $\epsilon \leftarrow f_2^* - 1$.
- 5 Stop when $\epsilon < z_2^*$ (or the MILP is infeasible).

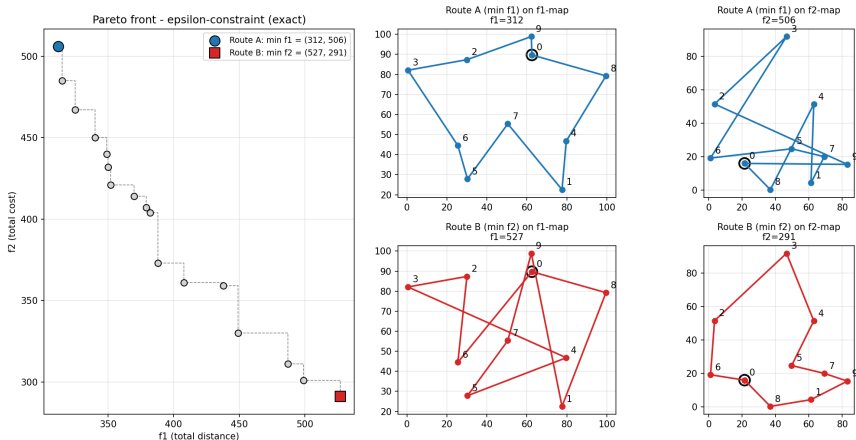
Numerical Results (HiGHS solver via SciPy)

Ideal $z^* = (312, 291)$ **Nadir** $z^{\text{nad}} = (527, 506)$

Iteration	ϵ	f_1	f_2
0 (lexicographic)	—	312	506
1	505	315	485
2	484	325	467
3	466	340	450
4	449	349	440
$\vdots$	$\vdots$	$\vdots$	$\vdots$
16	300	527	291

- **17 Pareto-optimal solutions** obtained with 16 ϵ -subproblems (plus 4 lexicographic MILPs), in about 5 s.
- The skip step avoids redundant subproblems: no repeated points.

Trade-off Between Distance and Cost



Route A (min f_1) is clean on the f_1 -map and tangled on the f_2 -map; Route B (min f_2) is the opposite. No single tour is optimal for both objectives.

① Foundational Value:

- The ϵ -constraint method is a rigorous scalarization tool capable of exploring non-convex Pareto regions.

② Modern Modifications:

- Slack variables (AUGMECON2) and adaptive early-exit mechanisms (S-AUGM) resolve classical issues of redundancy and weak efficiency.

③ Representation Capability:

- Grid Point Based Algorithms (GPBA) allow tailored generation focusing explicitly on coverage, uniformity, or cardinality.

Key References:

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